Investigation: SPRINGER SA (CVM code 10960), fiscal year 2016, as it was on 2017-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Máquinas, Equipamentos, Veículos e Peças (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

Document used

2016 DFP, version 1, filed on 2017-04-25; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2017-04-25).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=64391

Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: 0.822 (index)
- Alert by the study 1 rule: yes
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: 96 (n = 14)
- Percentile in group and year: 98 (n = 349)
- The company's own history: median −1.804; Tukey range (1.5 × IQR): from −7.380 to 2.173; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 0.325 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 3 (n = 17)
- Percentile in group and year: 4 (n = 425)
- The company's own history: median 0.786; Tukey range (1.5 × IQR): from 0.305 to 1.414; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

GMI, gross margin index
- Value: 0.326 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 15 (n = 17)
- Percentile in group and year: 6 (n = 438)
- The company's own history: median 1.244; Tukey range (1.5 × IQR): from −0.215 to 2.222; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

AQI, asset quality index
- Value: 1.516 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 79 (n = 17)
- Percentile in group and year: 95 (n = 426)
- The company's own history: median 1.264; Tukey range (1.5 × IQR): from −0.667 to 2.535; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

SGI, sales growth index
Value: 3.296 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 97 (n = 17)
Percentile in group and year: 98 (n = 440)
The company's own history: median 0.969; Tukey range (1.5 × IQR): from 0.137 to 1.967; value above the range (index; 5 fiscal years)
Confidence in the data: medium

DEPI, depreciation index
Value: 0.202 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 4 (n = 14)
Percentile in group and year: 2 (n = 366)
The company's own history: median 0.943; Tukey range (1.5 × IQR): from 0.766 to 1.135; value below the range (index; 5 fiscal years)
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 0.420 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 3 (n = 17)
Percentile in group and year: 3 (n = 433)
The company's own history: median 1.162; Tukey range (1.5 × IQR): from 0.361 to 1.899; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

TATA, total accruals to total assets
Value: 0.440 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 97 (n = 17)
Percentile in group and year: 98 (n = 452)
The company's own history: median −0.002; Tukey range (1.5 × IQR): from −0.859 to 0.727; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

LVGI, leverage index
Value: 1.132 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 79 (n = 17)
Percentile in group and year: 84 (n = 446)
The company's own history: median 0.814; Tukey range (1.5 × IQR): from 0.172 to 1.643; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

DPO jump
Value: −53.0 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 29.0 days)
Percentile in sector and year: 9 (n = 17)
Percentile in group and year: 6 (n = 410)
The company's own history: median −0.1 days; Tukey range (1.5 × IQR): from −13.1 days to 13.1 days; value below the range (5 fiscal years)
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix

(decision of 2026-11-07)

Implied interest rate minus average CDI
Value: zero denominator: average debt (no borrowings in t and t−1)
Alert by the study 1 rule: does not trigger alerts
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: —
Percentile in group and year: —
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: low
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: 0.408 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 97 (n = 17)
Percentile in group and year: 98 (n = 449)
The company's own history: median −0.002; Tukey range (1.5 × IQR): from −0.730 to 0.629; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

Financial risk (context only)

Altman Z" (financial risk, context only)
Value: 8.169 (index)
Components: X1 = 0.125; X2 = −0.194; X3 = −0.058; X4 = 7.972 (indices)
Percentile in sector and year: 97 (n = 18)
The company's own history: median 6.885; Tukey range (1.5 × IQR): from 1.445 to 10.530; value inside the range (index; 5 fiscal years)
- No cut-off zone: the available source does not define zones for Z"
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 3/9 (8 computable)
Signals:
  - Return on assets (ROA): 0
  - Operating cash flow on assets (CFO): 0
  - Change in ROA (ΔROA): 1
  - Cash above earnings (ACCRUAL): 0
  - Change in leverage (ΔLEVER): 0
  - Change in current ratio (ΔLIQUID): 0
  - No equity issuance (EQ_OFFER): —
  - Change in gross margin (ΔMARGIN): 1
  - Change in asset turnover (ΔTURN): 1
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement
- No long-term debt in either year: the change in leverage (ΔLEVER) is zero and the signal is 0 (project choice; the article only defines fell = 1, rose = 0, p. 8)

Changes

Restatement through the following year's comparative: no difference above the tolerance, or the next year's DFP did not exist yet on the date.

What was known on the date

- DFP 2011, v1: 2012-04-02
- DFP 2012, v1: 2013-03-27
- DFP 2013, v1: 2014-04-02
- DFP 2014, v1: 2015-03-31
- DFP 2015, v1: 2016-03-31
- FCA 2016, v1: 2016-03-31
- DFP 2016, v1: 2017-04-25

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 0 failed.

Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).
no opinion excerpt was found in the text
Auditor in the 2015 FCA: Confidor - Auditores Associados, Peppe Associados - Consultores & Auditores Independentes.
Auditor in the 2016 FCA: Peppe Associados - Consultores & Auditores Independentes.
Auditor in the 2017 FCA: Peppe Associados - Consultores & Auditores Independentes.
Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2015: Confidor - Auditores Associados -> Peppe Associados - Consultores & Auditores Independentes (transition: two firms in the 2015 FCA). The outgoing firm audited 5 consecutive fiscal years (2010 to 2014), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 38 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E).
Generated at 2026-11-11T00:46:03+00:00.

Today's information (outside the reference date)

Registry status: CANCELADA. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.
Name in today's registry: SPRINGER SA.